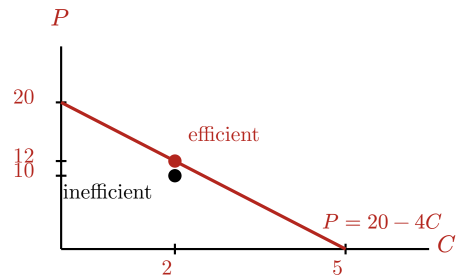
ECON 0100 | Vignette A1 | PPF | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see.

Colin Creevey can bake 20 cornish pasties (P) or 5 cauldron cakes (C) in one day. Set up Colin's PPF on an x, y graph with pasties (P) on the vertical and cakes (C) on the horizontal. Label the axes and the intercepts.

> Intercepts: 20 pasties on the vertical axis (a full day on pasties) and 5 cakes on the horizontal (a full day on cakes). The PPF is the line between them, $P = 20 - 4C$. The Q2 bundle $(2, 10)$ is marked below the frontier.



Q1 | Opportunity Cost

What is Colin's opportunity cost of producing 1 cake? Of 1 pasty?

Opportunity cost of 1C: 4 pasties

Opportunity cost of 1P: $\frac{1}{4}$ cake

> A full day is $20P$ or $5C$, so $5C = 20P$ and $1C = 4P$: one cake costs four pasties. Flip it for the other direction: $1P = \frac{1}{4}C$. The two opportunity costs are always reciprocals. The 4 is also the absolute slope of the PPF: pasties given up per extra cake.

Q2 | Feasibility

Suppose Colin bakes 10 pasties and 2 cakes in one day. Is this inefficient, efficient, or unattainable? Use a graph or algebra to justify your answer.

Inefficient, Efficient, Unattainable: Inefficient

> **Graph:** at $C = 2$ the frontier allows $P = 20 - 4(2) = 12$, and $(2, 10)$ sits below it, so inefficient: Colin could bake two more pasties without giving up a cake.
> **Algebra (time check):** 2 cakes take $\frac{2}{5}$ of the day and 10 pasties take $\frac{10}{20} = \frac{1}{2}$ of the day, so the bundle uses $\frac{2}{5} + \frac{1}{2} = \frac{9}{10}$ of a day. Under a full day is inefficient, exactly one day efficient, over one day unattainable. Either justification earns full credit.

Q3 | Next Best Alternative

In his free afternoon Colin would rather bake for the Gryffindor party than photograph the Quidditch match, and both of which he'd prefer over taking a nap. What is Colin's opportunity cost of baking?

Opportunity cost of baking before the invitation: Photographing the Quidditch match

Then Dennis invites him to Hogsmeade, which Colin likes more than photographing the match but less than baking. What is his opportunity cost of baking now?

Opportunity cost of baking after the invitation: The trip to Hogsmeade

> Opportunity cost is the single next-best alternative, not the sum of everything forgone. Before the invitation the ranking is baking > photographing > napping, so the cost of baking is the photography he gives up; the nap is not part of it, since he would not have napped anyway. Hogsmeade slots in between: baking > Hogsmeade > photographing > napping. Baking is still his best option, but it now costs more: the best thing he gives up is the Hogsmeade trip. The common mistake is "the match and Hogsmeade"; only one alternative can be next best.

ECON 0100 | Vignette A2 | Advantages | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see.

Colin Creevey can bake 20 cornish pasties (P) or 5 cauldron cakes (C) in one day. Katie Bell also bakes cornish pasties and cauldron cakes at a neighboring bakery. She can bake 15 pasties or 8 cakes in one day.

Q1 | Absolute Advantage

Set up a production table with both Colin and Katie's output per day. Who has the absolute advantage (AA) in pasties? In cakes?

AA in P: Colin

AA in C: Katie

Production per day	Pasties	Cakes
Colin	20	5
Katie	15	8

Absolute advantage is just "who makes more of it in a day": read straight down each column. Colin makes more pasties ($20 > 15$) and Katie makes more cakes ($8 > 5$).

Q2 | Comparative Advantage

Set up an opportunity cost table with Colin and Katie's opportunity costs for each good. Who has the comparative advantage (CA) in pasties? In cakes?

CA in P: Colin

CA in C: Katie

Each cell is "how much of the other good one unit costs," read off the production table: Colin's $20P = 5C$ gives $1C = 4P$ and $1P = \frac{1}{4}C$; Katie's $15P = 8C$ gives $1C = \frac{15}{8}P$ and $1P = \frac{8}{15}C$.

Opportunity cost	of 1 pasty	of 1 cake
Colin	$\frac{1}{4}$ C $= 0.25$ C	4 P
Katie	$\frac{8}{15}$ C ≈ 0.53 C	$\frac{15}{8}$ P $= 1.875$ P

Comparative advantage goes to the lower opportunity cost, again reading down each column. Pasties: $\frac{1}{4} < \frac{8}{15}$, so Colin. Cakes: $\frac{15}{8} < 4$, so Katie. Here AA and CA line up, which is why Q3 exists.

A useful check: the two entries in a row are reciprocals, so whoever has the lower cost in one good has the higher cost in the other. Nobody can have the comparative advantage in both.

Q3 | Better at Both

Suppose Katie buys a new oven and can now bake 25 pasties or 8 cakes in one day. Update both tables. Who has the comparative advantage in each good?

CA in P: Colin   CA in C: Katie

Production per day	Pasties	Cakes
Colin	20	5
Katie (new oven)	25	8

Opportunity cost	of 1 pasty	of 1 cake
Colin	$\frac{1}{4}$ C $= 0.25$ C	4 P
Katie (new oven)	$\frac{8}{25}$ C $= 0.32$ C	$\frac{25}{8}$ P $= 3.125$ P

Katie now makes more of both goods, so she has the absolute advantage in both. Her opportunity costs change: $25P = 8C$ gives $1C = \frac{25}{8}P$ and $1P = \frac{8}{25}C$; Colin's row is unchanged. Pasties: $\frac{1}{4} < \frac{8}{25}$, so Colin still has the comparative advantage in pasties. Cakes: $\frac{25}{8} < 4$, so Katie keeps the comparative advantage in cakes.

The point of the question: Katie is better at both, but **relatively** much better at cakes ($\frac{8}{5}$ as many cakes as Colin, only $\frac{25}{20}$ as many pasties), so her cheap good is cakes and Colin's is pasties. Absolute advantage in both never means comparative advantage in both.

ECON 0100 | Vignette A3 | Trade | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see.

Colin Creevey can bake 20 cornish pasties (P) or 5 cauldron cakes (C) in one day, and Katie Bell, back on her original oven, can bake 15 pasties or 8 cakes.

Q1 | Trade

Suppose Colin and Katie realize they can specialize and trade goods. After they specialize, what is a trade that would make them both better off?

1 C for 3 P *(any rate strictly between $\frac{15}{8}$ and 4 pasties per cake is correct)*

> Opportunity cost of a cake: Colin $4P$, Katie $\frac{15}{8}P$. Katie has the comparative advantage in cakes and Colin in pasties, so Katie bakes cakes all day ($8C$) and Colin bakes pasties all day ($20P$). A price for cakes works when it sits between the two opportunity costs, $\frac{15}{8} <$ price of 1 cake $<$ 4 pasties. Below 4, Colin pays less for a cake than baking one costs him; above $\frac{15}{8}$, Katie gets more for a cake than baking one costs her. Any number in the range is correct; 2 or 3 are the natural answers.
> **Worked check at $1C$ for $3P$**, trading 2 cakes for 6 pasties. Colin ends with $14P$ and $2C$; his own PPF only allows $12P$ at $C = 2$, so he consumes beyond his frontier. Katie ends with $6C$ and $6P$; her PPF allows only $3.75P$ at $C = 6$, so she does too. Common wrong answers: a rate outside the range, or specializing in the wrong goods.

Q2 | Accept or Reject

Katie offers Colin 1 cake for 5 pasties. Does Colin accept? Why or why not?

Accept or Reject: Reject

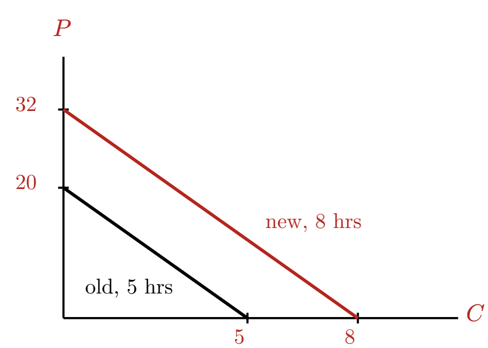
> Colin's own cost of a cake is 4 pasties. Paying 5 pasties for a cake is worse than baking it himself, so he rejects; the price is above the top of the range from Q1. Katie would love this trade ($5 > \frac{15}{8}$), but a trade needs both sides. The rule students should be able to state: each party accepts only prices better than their own opportunity cost.

Q3 | Changing Labor

It turns out Colin wants to add hours to his job, so he increases from 5 to 8 hours per day. Set up Colin's old and new PPF on the same graph. What is Colin's new opportunity cost of cake? Write a short description of how this increase in hours would impact the trade with Katie you found in Q1.

New opportunity cost of 1C: still 4 pasties

> Hours scale by $\frac{8}{5}$, and so does everything Colin can make: $20 \cdot \frac{8}{5} = 32$ pasties or $5 \cdot \frac{8}{5} = 8$ cakes. The new PPF is $P = 32 - 4C$, a parallel shift out of the old one. Both intercepts grow by the same factor, so the slope is unchanged: a cake still costs Colin 4 pasties. More time changes how much he can make, not the rate at which one good trades for the other inside his own bakery.



> **Effect on the trade:** since neither opportunity cost moved, the range of prices both accept is still $\frac{15}{8}$ to 4 pasties per cake, and the trade from Q1 still works at the same rate. What changes is the volume: Colin now has 32 pasties to work with instead of 20, so he can buy more of Katie's cakes. The size of the trade can grow; the terms do not have to.
> Grading note: the key idea is "opportunity cost unchanged, so the exchange rate range is unchanged." Credit "the trade gets bigger" or "Colin has more to trade" without insisting on a specific quantity.